



MINING DEALER BEST PRACTICE

Oil Filter Cutting Tool (Field Operations)

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

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1.0 Introduction

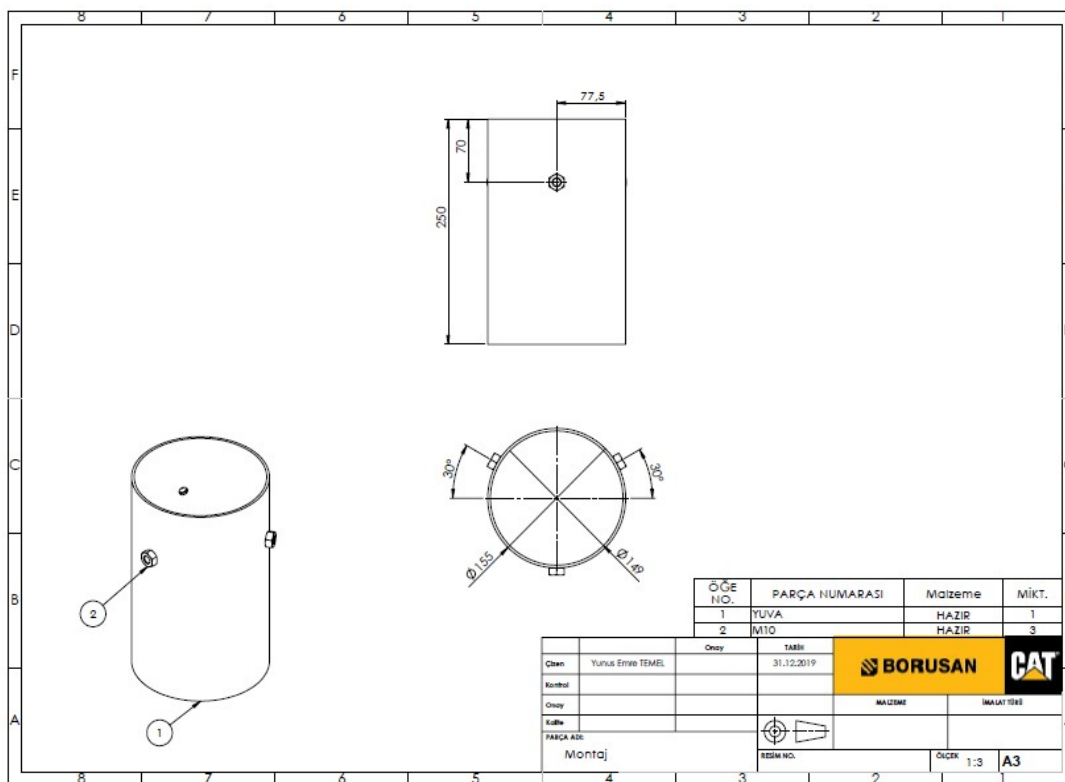
Oil filters must be inspected during maintenance in order to detect malfunctions of the components in advance and to prevent future malfunctions. It is necessary to remove the filter from the machine, cut the filter and check the elements in the filter.

Two best practices have been published in this area of work. One, the Oil Filter Cutting Bench, is a stand that holds the filter while it's being cut, and aids in the process of removing oil from the filter to check for abnormal elements. The other best practice, OHT Transmission & Hydraulic Filter Replacement Tool, is a ladder stand that allows the technician to safely and easily reach the filter and securely hold it while it's being cut. These best practices are more suited for shops and pits. However, there is not always an ability to check the filter in these areas. This portable filter cutting tool has been developed to provide easier service and maintenance processes in the field.

This tool ensures that the filters are cut without damaging the natural environment and the working environment. For example, most of the operations performed in the field use a variety of methods that cause work accidents, such as hand injuries, but this tool helps to reduce those accidents. This tool was also developed with a large oil sump that is aimed at eliminating the risk of oil leaking to the environment. This tool allows for easy portability as well, allowing machine inspections and maintenance to be performed in different locations, such as service vehicles.

2.0 Best Practice Description

A simple drawing of the filter cutting tool is given in the figure below. The chamber of the tool is made of 0.5 mm thick sheet metal. In addition, the fixing part is manufactured with an M14 stud.



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3.0 Implementation Steps

1. The filter is removed from the machine and placed in the hopper of the tool.
2. The filter is fixed at three points.



3. The fixed filter is cut with the help of the filter cutting tool and the process is completed.



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4.0 Benefits

- Portability that allows cutting in service vehicles or next to the machine during maintenance;
- Reduces work accidents by minimizing the risk of dropping the filter cutting tool;
- Eliminates oil leaking to the surrounding environment while cutting.

5.0 Resources Required

Item	Cost
Material	\$65
Labor	\$15
Total	\$80

6.0 Supporting Attachments / References

N/A.

7.0 Related Best Practices

- 1106-2.14-1036 - OHT Transmission and Hydraulic Filter Replacement Tool
- 1506-1.01-1381 - Oil Filter Cutting Bench

8.0 Acknowledgements

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